Chapter 5. The Tychonoff Theorem

Compactness of products, Stone–Čech compactification

Contents

5.1 The Tychonoff theorem **2**

5.2 The Stone–Čech compactification **3**

5.3 Selected exercises **4**

5.1 The Tychonoff theorem

Motivation. Arbitrary products of compact spaces are compact; the proof uses Zorn's lemma, and the theorem is equivalent to the axiom of choice.

We prove compactness via the finite intersection property (Theorem 3.23): a space is compact iff every collection $\mathcal{A}$ of subsets with the finite intersection property (FIP) satisfies $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Lemma 5.1 (37.1)

Let $\mathcal{A}$ be a collection of subsets of X with the FIP. Then there is a collection $\mathcal{D} \supset \mathcal{A}$ with the FIP such that no collection properly containing $\mathcal{D}$ has the FIP ($\mathcal{D}$ is *maximal* with respect to the FIP).

Proof. Let $\mathbb{A}$ be the set of all collections $\mathcal{B} \supset \mathcal{A}$ of subsets of X with the FIP, strictly partially ordered by proper inclusion. If $\mathbb{B} \subset \mathbb{A}$ is a chain, $\mathcal{C} = \bigcup_{\mathcal{B}\in\mathbb{B}} \mathcal{B}$ has the FIP: finitely many $C_1, \ldots, C_n \in \mathcal{C}$ lie in $\mathcal{B}_1, \ldots, \mathcal{B}_n$, all contained in the largest $\mathcal{B}_k$ (chain), which has the FIP. So $\mathcal{C}$ is an upper bound, and Zorn's lemma (Theorem 1.29) gives a maximal element. ■

Lemma 5.2 (37.2)

Let $\mathcal{D}$ be maximal with respect to the FIP.
(a) Every finite intersection of elements of $\mathcal{D}$ is in $\mathcal{D}$.
(b) If $A \subset X$ intersects every element of $\mathcal{D}$, then $A \in \mathcal{D}$.

Proof. (a) Let B be a finite intersection of elements of $\mathcal{D}$; $\mathcal{E} = \mathcal{D} \cup \{B\}$ has the FIP (a finite intersection of members of $\mathcal{E}$ is a finite intersection of members of $\mathcal{D}$), so $\mathcal{E} = \mathcal{D}$ by maximality.

(b) $\mathcal{E} = \mathcal{D} \cup \{A\}$: a finite intersection $A \cap D_1 \cap \cdots \cap D_n$ is nonempty since $D_1 \cap \cdots \cap D_n \in \mathcal{D}$ by (a) and A meets it. So $A \in \mathcal{D}$. ■

Theorem 5.3 (37.3, Tychonoff theorem)

An arbitrary product of compact spaces is compact in the product topology.

Proof. Let $X = \prod_{\alpha\in J} X_\alpha$ and $\mathcal{A}$ a collection of subsets of X with the FIP; extend it to a maximal $\mathcal{D}$ (Lemma 5.1). It suffices to find $x \in \bigcap_{D\in\mathcal{D}} \overline{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so choose $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$, and let $x = (x_\alpha)$.

Step 1. if U_β is a neighbourhood of x_β, then $\pi_\beta^{-1}(U_\beta)$ meets every $D \in \mathcal{D}$ (since U_β meets $\pi_\beta(D)$, as $x_\beta \in \overline{\pi_\beta(D)}$), so $\pi_\beta^{-1}(U_\beta) \in \mathcal{D}$ by Lemma 5.2(b).

Step 2. every basis element containing x is a finite intersection of such subbasis elements, hence lies in $\mathcal{D}$ (Lemma 5.2(a)), hence meets every element of $\mathcal{D}$ (FIP). So $x \in \overline{D}$ for all $D \in \mathcal{D}$. ■

The theorem fails for the box topology ($[0,1]^\omega$ in the box topology is not compact), and it is equivalent to the axiom of choice.

Example 5.1

$[0,1]^\omega$ is a compact metrizable space; every regular second countable space imbeds in it (proof of Theorem 4.12). $[0,1]^J$ for uncountable J is compact Hausdorff but not metrizable: it is not even first countable (a countable family of basis neighbourhoods of a point restricts only countably many coordinates).

5.2 The Stone–Čech compactification

Motivation. Every completely regular space has a largest compactification βX, the closure of X embedded in a product of intervals, to which every bounded continuous function extends.

A **compactification** of X is a compact Hausdorff space Y containing X as a subspace with $\overline{X} = Y$; two are **equivalent** if there is a homeomorphism between them that is the identity on X. If X has a compactification, it is completely regular (a subspace of a compact Hausdorff, hence normal, space).

Lemma 5.4 (38.1)

If $h : X \to Z$ is an imbedding in a compact Hausdorff space Z, there is a compactification Y of X and an imbedding $H : Y \to Z$ equal to h on X; Y is unique up to equivalence (Y is the compactification **induced** by h).

Proof. Let $X_0 = h(X)$ and $Y_0 = \overline{X_0}$, a compact Hausdorff space. Form $Y = X \sqcup (Y_0 - X_0)$ with the bijection $H : Y \to Y_0$ (h on X, identity elsewhere) and give Y the topology making H a homeomorphism. Then X is a subspace of Y, dense since X_0 is dense in Y_0. ■

Theorem 5.5 (38.2)

Every completely regular space X has a compactification Y such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous $Y \to \mathbb{R}$.

Proof. Let $\{f_\alpha\}_{\alpha \in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset I_\alpha$, where $I_\alpha = [\inf f_\alpha, \sup f_\alpha]$. $h = (f_\alpha) : X \to Z = \prod I_\alpha$ is an imbedding (Theorem 4.13, using complete regularity), and Z is compact Hausdorff (Tychonoff). Let Y be the induced compactification with $H : Y \to Z$. For $f = f_\beta$, $\pi_\beta \circ H$ is a continuous extension. Uniqueness follows from the next lemma. ■

Lemma 5.6 (38.3)

If $A \subset X$ and $f : A \to Z$ is continuous with Z Hausdorff, there is at most one continuous extension $g : \overline{A} \to Z$.

Proof. If $g \neq g'$ at x, separate $g(x), g'(x)$ by disjoint U, U'; the neighbourhood $g^{-1}(U) \cap g'^{-1}(U')$ of x contains a point $y \in A$, but $g(y) = g'(y) = f(y)$ lies in $U \cap U'$. ■

Theorem 5.7 (38.4)

If Y is a compactification of the completely regular X with the extension property of Theorem 5.5, every continuous map f of X into a compact Hausdorff space C extends uniquely to a continuous $g : Y \to C$.

Proof. C is completely regular, so imbeds in some $[0,1]^J$ (Corollary 4.14); regard $C \subset [0,1]^J$. Each coordinate $\pi_\alpha \circ f$ extends to $g_\alpha : Y \to [0,1]$; $g = (g_\alpha) : Y \to [0,1]^J$ is continuous. Since $g(X) = f(X) \subset C$ and X is dense, $g(Y) \subset \overline{g(X)} \subset C$ (C is closed).

Uniqueness: Lemma 5.6. ■

Theorem 5.8 (38.5)

Two compactifications Y_1, Y_2 of X with the extension property of Theorem 5.5 are equivalent.

Proof. Extend the inclusion $j_2 : X \to Y_2$ to $f_2 : Y_1 \to Y_2$ and j_1 to $f_1 : Y_2 \to Y_1$ (Theorem 5.7). $f_1 \circ f_2 : Y_1 \to Y_1$ extends the identity of X, as does id_{Y_1}; by Lemma 5.6, $f_1 \circ f_2 = \mathrm{id}$. Similarly $f_2 \circ f_1 = \mathrm{id}$. ∎

This compactification is the **Stone–Čech compactification** $\beta(X)$. It is characterized by the universal property of Theorem 5.7.

Example 5.2

(a) The one-point compactification of $(0,1)$ is the circle and $[0,1]$ is a two-point compactification; neither is $\beta(0,1)$, since $\sin(1/x)$ extends to $\beta(0,1)$.
(b) $\beta(\mathbb{Z}_+)$ is not sequentially compact: $\mathbb{Z}_+$ has no convergent subsequence in $\beta\mathbb{Z}_+$ (separate even and odd subsequences by extending functions on $\mathbb{Z}_+$).
(c) $\beta(S_\Omega) = \overline{S}_\Omega$, because every continuous real function on S_Ω is eventually constant.

5.3 Selected exercises

Motivation. Exercises from Munkres on the countable intersection criterion, maximality of $\beta(X)$, the fact about S_Ω used in Example 5.2(c), and connectedness of βX.

Exercise 5.1 (§37, Ex. 2; Lindelöf spaces)

A collection $\mathcal{A}$ of subsets of X has the **countable intersection property** if every countable intersection of elements of $\mathcal{A}$ is nonempty. Show that X is Lindelöf iff for every collection $\mathcal{A}$ with the countable intersection property, $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Solution.

($\Rightarrow$) Let X be Lindelöf, $\mathcal{A}$ with the countable intersection property, and suppose $\bigcap_{A\in\mathcal{A}} \overline{A} = \emptyset$. Then $\{X - \overline{A} : A \in \mathcal{A}\}$ is an open covering of X, so countably many $X - \overline{A_n}$ cover X, i.e. $\bigcap_n \overline{A_n} = \emptyset$. But $\bigcap_n A_n \subset \bigcap_n \overline{A_n}$ is nonempty, a contradiction.

($\Leftarrow$) Let $\mathcal{U}$ be an open covering with no countable subcovering, and $\mathcal{A} = \{X - U : U \in \mathcal{U}\}$. For $U_1, U_2, \cdots \in \mathcal{U}$, $\bigcap_n (X - U_n) = X - \bigcup_n U_n \neq \emptyset$, so $\mathcal{A}$ has the countable intersection property. Its members are closed, and $\bigcap_{U\in\mathcal{U}}(X - U) = X - \bigcup \mathcal{U} = \emptyset$, contradicting the hypothesis. So X is Lindelöf. ∎

Exercise 5.2 (§38, Ex. 4; $\beta(X)$ is the largest compactification)

Let Y be any compactification of the completely regular space X. Show that there is a continuous, closed, surjective map $g : \beta(X) \to Y$ that is the identity on X.

Solution.

Step 1. The inclusion $j : X \to Y$ is a continuous map into a compact Hausdorff space, so by Theorem 5.7 it extends to a continuous $g : \beta(X) \to Y$, and $g = j$ is the identity on X.

Step 2. (g is closed.) A closed $C \subset \beta(X)$ is compact (Theorem 3.16), so $g(C)$ is compact (Theorem 3.19), hence closed in the Hausdorff space Y (Theorem 3.17).

Step 3. (g is onto.) By Step 2, $g(\beta(X))$ is closed in Y; it contains X, which is dense in Y. So $g(\beta(X)) = Y$.

A closed continuous surjection is a quotient map, so every compactification of X is equivalent to a quotient space of $\beta(X)$. ■

Exercise 5.3 (§38, Ex. 5(a); functions on S_Ω)

Show that every continuous $f : S_\Omega \to \mathbb{R}$ is eventually constant: there is $\alpha \in S_\Omega$ with $f(\beta) = f(\alpha)$ for all $\beta > \alpha$.

Solution.

Step 1. (Increasing sequences converge.) Let $\alpha_1 < \alpha_2 < \cdots$ in S_Ω. By Theorem 1.27 the set of upper bounds of $\{\alpha_n\}$ in S_Ω is nonempty; let b be its least element. If $c < b$, then c is not an upper bound, so $\alpha_N > c$ for some N, and $\alpha_n \in (c, b]$ for $n \geq N$. Since $b > \alpha_1$ is not the least element of S_Ω, every neighbourhood of b contains such an interval $(c, b]$ with $c < b$; hence $\alpha_n \to b$, and by continuity $f(\alpha_n) \to f(b)$.

Step 2. (Small oscillation eventually.) Fix $\epsilon > 0$. We claim there is α with $|f(\beta) - f(\alpha)| < \epsilon$ for all $\beta > \alpha$. If not, then for every α there is $\beta > \alpha$ with $|f(\beta) - f(\alpha)| \geq \epsilon$. Starting from any α_1, we obtain $\alpha_1 < \alpha_2 < \cdots$ with $|f(\alpha_{n+1}) - f(\alpha_n)| \geq \epsilon$ for all n. Then $(f(\alpha_n))$ is not Cauchy, contradicting Step 1.

Step 3. (Conclusion.) For each n choose α_n as in Step 2 with $\epsilon = 1/n$, and by Theorem 1.27 an upper bound α of $\{\alpha_n\}$. For $\beta > \alpha$ and each n, both β and α are $\geq \alpha_n$, so

$$|f(\beta) - f(\alpha)| \leq |f(\beta) - f(\alpha_n)| + |f(\alpha_n) - f(\alpha)| < \tfrac{2}{n}$$

(the second term is 0 if $\alpha = \alpha_n$). Letting $n \to \infty$, $f(\beta) = f(\alpha)$. ■

Exercise 5.4 (§38, Ex. 6; connectedness of $\beta(X)$)

Let X be completely regular. Show that X is connected iff $\beta(X)$ is connected.

Solution.

($\Rightarrow$) X is connected and dense in $\beta(X)$, and the closure of a connected set is connected, so $\beta(X) = \overline{X}$ is connected.

($\Leftarrow$) Suppose $X = A \cup B$ is a separation. Then $f = 0$ on A, $f = 1$ on B is continuous and bounded (A, B are open), so it extends to a continuous $g : \beta(X) \to \mathbb{R}$ (Theorem 5.5). Since X is dense, $g(\beta(X)) = g(\overline{X}) \subset \overline{g(X)} = \{0, 1\}$, and both values occur. So $g^{-1}(0)$ and $g^{-1}(1)$ form a separation of $\beta(X)$, a contradiction. ■